Homework 4: Qualitative Analysis III — Mixtures & Equations

Paper track. Pairs with Home Lab 4. Do this on your own. Calculator OK.

Part A: Mixture logic (2 pts each)

For each, say **single powder** or **mixture**, and name the component(s):

1. Fizzes strongly, dissolves fully, pH 9.
2. Fizzes weakly, most of it won't dissolve, iodine-negative, gritty residue.
3. Half dissolves; dissolved part is acidic (pH 3); undissolved part turns iodine blue-black.
4. Dissolves fully; two different crystal shapes visible under the lens; neutral; iodine-negative.

Part B: Balancing (1 pt each)

1. $\_\_ \; H_2 + \_\_ \; O_2 \rightarrow \_\_ \; H_2O$
2. $\_\_ \; Na + \_\_ \; Cl_2 \rightarrow \_\_ \; NaCl$
3. $\_\_ \; Fe + \_\_ \; O_2 \rightarrow \_\_ \; Fe_2O_3$
4. $\_\_ \; Mg + \_\_ \; HCl \rightarrow \_\_ \; MgCl_2 + \_\_ \; H_2$
5. $\_\_ \; C_3H_8 + \_\_ \; O_2 \rightarrow \_\_ \; CO_2 + \_\_ \; H_2O$
6. $\_\_ \; NaHCO_3 + \_\_ \; H_3C_6H_5O_7 \rightarrow \_\_ \; CO_2 + \_\_ \; H_2O + \_\_ \; Na_3C_6H_5O_7$

Part C: Worked example — copy the method (6 pts)

Solved for you: 12 g of baking soda ($NaHCO_3$, 84 g/mol) reacts with excess vinegar. How many grams of CO_2 (44 g/mol) are produced? $NaHCO_3 + CH_3COOH \rightarrow NaC_2H_3O_2 + H_2O + CO_2$

- moles $NaHCO_3$ = 12 ÷ 84 = 0.143 mol
- ratio $NaHCO_3$: CO_2 is 1 : 1 → 0.143 mol CO_2
- mass CO_2 = 0.143 × 44 = **6.3 g**

Now you do it: 21 g of baking soda reacts with excess vinegar. How many grams of CO_2 are produced? Show every step.

Part D: Limiting reactant (4 pts)

You mix 0.40 mol of $NaHCO_3$ (coefficient 3) with 0.10 mol of citric acid (coefficient 1), following $3\ NaHCO_3 + H_3C_6H_5O_7 \rightarrow 3\ CO_2 + 3\ H_2O + Na_3C_6H_5O_7$.

1. Which is the limiting reactant? Show the division.
2. How many moles of CO_2 form?

Answer Key

Part A: 1. Single — baking soda. 2. Mixture — a carbonate (baking soda) + sand (SiO_2, gritty, insoluble). 3. Mixture — citric acid (soluble, acidic) + corn starch (insoluble, iodine +). 4. Mixture — e.g. table salt + sugar (both soluble, neutral, iodine-negative, different crystal shapes).

Part B: 1) 2,1,2 2) 2,1,2 3) 4,3,2 4) 1,2,1,1 5) 1,5,3,4 6) 3,1,3,3,1

Part C: moles $NaHCO_3$ = 21 ÷ 84 = 0.25 mol; ratio 1:1 → 0.25 mol CO_2; mass = 0.25 × 44 = **11 g CO_2.**

Part D:

1. $NaHCO_3$: 0.40 ÷ 3 = 0.133. Citric acid: 0.10 ÷ 1 = 0.10. 0.10 is smaller → **citric acid is limiting**.
2. CO_2 : citric acid ratio is 3 : 1 → 0.10 × 3 = **0.30 mol CO_2.**